

DATA STRUCTURES OVERVIEW

SOURCES OF DATA

- 1 **From the program itself:** Data written directly in source code (e.g. status messages)
- 2 **From the UI:** Data input from the user or data written in DOM (e.g. tasks in todo app)
- 3 **From external sources:** Data fetched for example from web API (e.g. recipe objects)

Collection of data

Data structure

SIMPLE LIST?

KEY/VALUE PAIRS?

Arrays or Sets

Objects or Maps

OTHER BUILT-IN:

- WeakMap
- WeakSet

NON-BUILT IN:

- Stacks
- Queues
- Linked lists
- Trees
- Hash tables

Application
Programming
Interface

"Object"

Array

```
{
  "count": 3,
  "recipes": [
    {
      "publisher": "101 Cookbooks",
      "title": "Best Pizza Dough Ever",
      "source_url": "http://www.101cookbooks.com/archiv",
      "recipe_id": "47746",
      "image_url": "http://forkify-api.herokuapp.com/im",
      "social_rank": 100,
      "publisher_url": "http://www.101cookbooks.com"
    },
    {
      "publisher": "The Pioneer Woman",
      "title": "Deep Dish Fruit Pizza",
      "source_url": "http://thepioneerwoman.com/cooking",
      "recipe_id": "46956",
      "image_url": "http://forkify-api.herokuapp.com/im",
      "social_rank": 100,
      "publisher_url": "http://thepioneerwoman.com"
    },
    {
      "publisher": "Closet Cooking",
      "title": "Pizza Dip",
      "source_url": "http://www.closetcooking.com/2011/",
      "recipe_id": "35477",
      "image_url": "http://forkify-api.herokuapp.com/im",
      "social_rank": 99.999999999999994,
      "publisher_url": "http://closetcooking.com"
    }
  ]
}
```

Keys allow us to
describe values

JSON data format example

ARRAYS VS. SETS AND OBJECTS VS. MAPS

ARRAYS

VS.

SETS

```
tasks = ['Code', 'Eat', 'Code'];  
// ["Code", "Eat", "Code"]
```

```
tasks = new Set(['Code', 'Eat', 'Code']);  
// {"Code", "Eat"}
```

- 👉 Use when you need **ordered** list of values (might contain duplicates)
- 👉 Use when you need to **manipulate** data

- 👉 Use when you need to work with **unique** values
- 👉 Use when **high-performance** is *really* important
- 👉 Use to **remove duplicates** from arrays

OBJECTS

VS.

MAPS

```
task = {  
  task: 'Code',  
  date: 'today',  
  repeat: true  
};
```

```
task = new Map([  
  ['task', 'Code'],  
  ['date', 'today'],  
  [false, 'Start coding!']  
]);
```

- 👉 More “traditional” key/value store (“abused” objects)
- 👉 Easier to write and access values with `.` and `[]`
- 👉 Use when you need to include **functions** (methods)
- 👉 Use when working with JSON (can convert to map)

- 👉 Better performance
- 👉 Keys can have **any** data type
- 👉 Easy to iterate
- 👉 Easy to compute size
- 👉 Use when you simply need to map key to values
- 👉 Use when you need keys that are **not** strings